

Probabilistic method

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Discovery is a child's privilege. I mean the small child, the child who is not afraid to be wrong, to look silly, to not be serious, and to act differently from everyone else. ¹

1 Probability Itself

Proposition 1.1

If the probability that an outcome is in a specific subset is non-zero, then that subset must be non-empty.

The method of application is as follows: to prove the existence of an object with a desired property, we sometimes show that if we choose an object at random from a larger set, it has the desired property with a positive probability.

An important tool in calculating probabilities is conditional probability, which allows us to break down complex problems into simpler ones. Conditional probability is defined as follows:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

An important property that makes conditional probability useful is the law of total probability. If the events $B_1, B_2, \dots, B_k$ form a partition of the sample space, we have:

$$P(A) = \sum_{i=1}^k P(A|B_i)P(B_i)$$

Now, suppose we are playing a game and, for example, if the die roll is an odd number, we win a certain amount of money. A natural question is, if we repeat this game many times, how much money will we win on average? This topic, aside from its use in solving mathematical problems, is also important in the real world. For example, suppose you have to pay an entry fee to participate in a game. If your average winnings are much less than the entry fee, is it rational to participate?

The mathematical way to model the above problem is through the concept of a random variable. Suppose we have a sample space where each outcome occurs with a specific probability. A random variable is a function from this sample space to the real numbers.

Example 1.2

Suppose our sample space consists of all possible outcomes of tossing a coin 100 times. Let X be a function that assigns to each outcome the number of heads. This function is an example of a random variable. We can also define random variables X_i such that $X_i = 1$ if the i -th toss is heads, and $X_i = 0$ otherwise. In this case, we have $X = \sum_{i=1}^{100} X_i$.

¹Grothendieck

Example 1.3

Consider the set of all vertices of a graph, and assume that the probability of selecting each vertex is uniform. The variable Y that assigns to each vertex its degree is an example of a random variable.

Now let's return to the problem of the average money won. Mathematically, this concept is called expected value.

Definition 1.4. If $X : M \rightarrow \mathbb{R}$ is a random variable on a sample space M , we define the expected value of X as follows:

$$E[X] = \sum_{m \in M} p(m)X(m)$$

Expected value has a simple but useful property, linearity, which can be easily verified: $E[X + Y] = E[X] + E[Y]$.

Problem 1.5. Ali tossed a coin 11 times and Reza 10 times. What is the probability that Ali has more heads than Reza?

Problem 1.6 (Komal A800). In a finite, simple, connected graph G , we play the following game: initially we color all the vertices with a different color. In each step we choose a vertex randomly (with uniform distribution), and then choose one of its neighbors randomly (also with uniform distribution), and color it to the same color as the originally chosen vertex (if the two chosen vertices already have the same color, we do nothing). The game ends when all the vertices have the same color. Knowing graph G find the probability for each vertex that the game ends with all vertices having the same color as the chosen vertex.

Problem 1.7. Ameneh and Baran are playing tennis. When Ameneh serves, she wins the point with 60% probability. When Baran serves, Ameneh wins with 50% probability. The first person to win 12 games wins the match. Which serving method is better for Ameneh: alternating serves, or the winner of a point serves the next one?

Problem 1.8. Pick $2n \geq 4$ points uniformly at random in the interval $[0, 1]$, say, $X_1, X_2, \dots, X_{2n}$ and, for $1 \leq i \leq n$, let J_i be the closed interval with end-points X_{2i-1} and X_{2i} . Then the probability that there is an interval J_i that meets all other intervals is $2/3$, independently of n .

2 First momentum

Expected value is a type of weighted average and therefore satisfies the following simple property, which is useful in solving some problems.

Proposition 2.1

There always exists an outcome m in the sample space such that $X(m) \geq E[X]$.

Example 2.2

In a 100×100 grid, the numbers from 1 to 100 are written such that each number appears 100 times. Prove that there is a row or a column that contains at least 10 distinct numbers.

Solution. Let's choose one of the 200 lines (100 rows and 100 columns) uniformly at random. Let X be the random variable representing the number of distinct numbers in the chosen line. By Proposition 2.1, there must exist a line with at least $E[X]$ distinct numbers. We will show that $E[X] \geq 10$.

Let X_i be the indicator random variable that the number i (for $i = 1, \dots, 100$) is present in the chosen line. Then $X = \sum_{i=1}^{100} X_i$. By linearity of expectation, $E[X] = \sum_{i=1}^{100} E[X_i]$.

$E[X_i]$ is the probability that the number i is in a randomly chosen line. We can calculate the sum of the number of distinct elements over all lines using double counting:

$$\sum_{\text{line } L} (\text{number of distinct elements in } L) = \sum_L \sum_{i=1}^{100} 1_{i \in L} = \sum_{i=1}^{100} \sum_L 1_{i \in L}$$

The inner sum $\sum_L 1_{i \in L}$ counts the number of lines containing the number i . Let r_i be the number of rows containing i and c_i be the number of columns containing i . The total number of lines containing i is $r_i + c_i$. Since the number i appears 100 times, its occurrences must be spread across at least r_i rows and c_i columns, so $r_i c_i \geq 100$. By the AM-GM inequality, $r_i + c_i \geq 2\sqrt{r_i c_i} \geq 2\sqrt{100} = 20$.

The sum is therefore $\sum_{i=1}^{100} (r_i + c_i) \geq \sum_{i=1}^{100} 20 = 2000$. The expected value is the total count divided by the number of lines (200):

$$E[X] = \frac{1}{200} \sum_L (\text{number of distinct elements in } L) \geq \frac{2000}{200} = 10.$$

Since $E[X] \geq 10$, there must be at least one line with 10 or more distinct numbers. $\square$

Now consider a harder problem.

Example 2.3 (Russia 1996)

The Russian Duma has 1600 members who form 16000 committees of 80 members each. Prove that there exist two committees that have at least four members in common.

Solution. Consider a random pair of distinct committees, chosen uniformly from all possible pairs. Let X be the random variable for the number of common members in these two committees. We want to show that there exists a pair for which $X \geq 4$. By the probabilistic method, it suffices to show that $E[X] > 3$.

Let X_i be the indicator random variable that person i (for $i = 1, \dots, 1600$) is a member of both chosen committees. It is clear that

$$X = \sum_{i=1}^{1600} X_i$$

By linearity of expectation, $E[X] = \sum_{i=1}^{1600} E[X_i]$. $E[X_i]$ is the probability that person i is in both committees. If person i is a member of n_i committees, the number of pairs of committees containing person i is $\binom{n_i}{2}$. The total number of pairs of committees is $\binom{16000}{2}$. So,

$$E[X_i] = \frac{\binom{n_i}{2}}{\binom{16000}{2}}$$

We know that the sum of the n_i 's, $\sum_{i=1}^{1600} n_i$, is the total number of membership slots across all committees, which is 16000×80 . So,

$$E[X] = \sum_{i=1}^{1600} \frac{\binom{n_i}{2}}{\binom{16000}{2}} = \frac{1}{\binom{16000}{2}} \sum_{i=1}^{1600} \binom{n_i}{2}$$

The function $f(x) = \binom{x}{2} = \frac{x(x-1)}{2}$ is convex. By Jensen's inequality, the sum $\sum \binom{n_i}{2}$ is minimized when the n_i are as equal as possible. The average value of n_i is $\bar{n} = \frac{\sum n_i}{1600} = \frac{16000 \times 80}{1600} = 800$. Therefore,

$$E[X] \geq \frac{1}{\binom{16000}{2}} \cdot 1600 \cdot \binom{800}{2} = \frac{1600 \cdot \frac{800 \cdot 799}{2}}{\frac{16000 \cdot 15999}{2}} = \frac{1600 \cdot 800 \cdot 799}{16000 \cdot 15999} = \frac{80 \cdot 799}{15999} = \frac{63920}{15999} \approx 3.995 > 3$$

Since $E[X] > 3$, there must exist at least one pair of committees with more than 3 common members, which means at least 4. $\square$

Now try to solve the following two problems with this idea.

Problem 2.4 (Iran TST 2008). 779 teams participate in a volleyball tournament where every team plays every other team exactly once. Prove that there exist two disjoint seven-member sets of teams, A and B , such that every team in A has defeated every team in B .

Problem 2.5 (IMO Shortlist 1999). Let A be a set of n residues modulo n^2 . Prove that there exists a set B of n residues modulo n^2 such that the set of sums $\{a + b \pmod{n^2} \mid a \in A, b \in B\}$ has size at least $n^2/2$.

Now let's move to problems where a probabilistic perspective provides even more help.

Example 2.6 (Russia 1999)

In a group of boys and men, every boy has shaken hands with at least one man, and every man has shaken hands with at least one boy. No two boys and no two men have shaken hands with each other. Prove that there is a subset containing at least half of the total people, such that every boy in this subset has shaken hands with an odd number of men in this subset.

Solution. Let's form a random subset as follows: select each man with probability $1/2$, independently. Then, add to the subset every boy who has shaken hands with an odd number of the selected men. Let S be the resulting set. We claim that every person is in this set with probability $1/2$. For the men, this is clear by construction. For a boy b , let's say he has shaken hands with $d_b \geq 1$ men. The probability that he is included in the set is the probability that an odd number of these d_b men are selected.

This is equivalent to the probability of getting an odd number of heads in d_b tosses of a fair coin. A simple combinatorial argument shows that for $d_b \geq 1$, the number of ways to choose an odd-sized subset from a set of size d_b is $\sum_{k \text{ odd}} \binom{d_b}{k} = 2^{d_b-1}$. The total number of ways to choose a subset of men he knows is 2^{d_b} . So the probability is $\frac{2^{d_b-1}}{2^{d_b}} = \frac{1}{2}$.

Thus, every person is in the set S with probability $1/2$. The expected size of S is the sum of these probabilities, which is half the total number of people. By Proposition 2.1, there must exist such a subset with at least this expected size. $\square$

Example 2.7

Suppose in a group of n people, friendship is a mutual relationship, and the average number of friends a person has is d . Prove that for any k such that $1 \leq k \leq d$, one can choose a subset of at least $\frac{kn}{d+1}$ members that does not contain a $(k+1)$ -clique (i.e., no $k+1$ people in the subset are all pairwise friends).

Solution. Model the problem with a graph where vertices are people and edges represent friendships. First, choose a random permutation of the vertices. Construct a set S greedily by iterating through the vertices in the order of the permutation. Add a vertex to S if it does not form a $(k+1)$ -clique with the vertices already in S . As in previous problems, let X be the size of the resulting set S , and let X_v be the indicator that vertex v is in S .

Consider a vertex v and its neighborhood $N(v)$. Let d_v be the degree of v . Consider the set $\{v\} \cup N(v)$ of size $d_v + 1$. If v appears among the first k vertices of this set in the random permutation, then when we consider v , at most $k-1$ of its neighbors can have been placed in S . Thus, v and its neighbors in S cannot

form a clique of size $k + 1$. So, adding v to S will not create a $(k + 1)$ -clique where v is a member. The probability of this is at least $\frac{k}{d_v + 1}$. Therefore, $P(v \in S) \geq \frac{k}{d_v + 1}$. By linearity of expectation,

$$E[X] = \sum_{v \in V} P(v \in S) \geq \sum_{v \in V} \frac{k}{d_v + 1} = k \sum_{v \in V} \frac{1}{d_v + 1}$$

By Jensen's inequality, since $f(x) = 1/(x + 1)$ is convex for $x \geq 0$, and the average degree is d , we have:

$$E[X] \geq k \cdot n \cdot \frac{1}{d + 1} = \frac{nk}{d + 1}$$

Thus, there exists a set S with size at least $\frac{nk}{d + 1}$. $\square$

Problem 2.8 (IMO Shortlist 1986). Let n, k be positive integers. Let S be a set of n points in the plane such that no three points are collinear, and for every point $P \in S$, there are at least k other points in S at the same distance from P . Prove that $k \leq \frac{1}{2} + \sqrt{2n}$.

Problem 2.9 (IMO Shortlist 1989). Let n be a positive integer. We have n points in the plane, and the set of distances between pairs of points has size 2. Let r be the number of red segments among them. If $m = \lceil \frac{2r}{n} \rceil$, prove that there is a red path of length m whose segment lengths are strictly increasing.

Problem 2.10. Prove that if the edges of a complete graph on n vertices are covered by a collection of complete bipartite graphs, the sum of the number of vertices in these bipartite graphs is at least $n \log_2 n$.

Example 2.11 (USAMO 2012, problem 2)

A circle is divided into 432 congruent arcs by 432 points. The points are colored in four colors such that some 108 points are colored Red, some 108 points are colored Green, some 108 points are colored Blue, and the remaining 108 points are colored Yellow. Prove that one can choose three points of each color in such a way that at the four triangles formed by the chosen points of the same color are congruent.

3 When you rule at the impossible

Let's start with an application of the most basic principle: the sum of the probabilities of all possible outcomes is one.

Example 3.1 (Leningrad 1987)

Let $A_1, A_2, \dots, A_s$ be subsets of $\{1, 2, \dots, m\}$ such that no set is a subset of another. Let the size of A_i be a_i . Prove that

$$\sum_{i=1}^s \frac{1}{\binom{m}{a_i}} \leq 1$$

Solution. To prove this, choose a random permutation of the set $\{1, 2, \dots, m\}$. For each i , let E_i be the event that the set of the first a_i elements of the permutation is A_i . By symmetry, the probability of this event is

$$P(E_i) = \frac{1}{\binom{m}{a_i}}$$

Note that the events E_i are mutually exclusive. If E_i and E_j both occurred for $i \neq j$, it would mean that the set of the first a_i elements of the permutation is A_i and the set of the first a_j elements is A_j . If $a_i = a_j$,

then $A_i = A_j$, which is not the case for distinct sets. If $a_i < a_j$, then A_i would be a subset of A_j , which is forbidden by the problem statement. Thus, at most one of these events can occur. Therefore, the sum of their probabilities is at most one, which is the desired inequality. $\square$

Problem 3.2. The vertices of a tetrahedron P lie on the surface of a sphere. Show that if more than $\frac{4}{9}$ of the surface of the sphere is colored, there exists a rotation of P such that all its vertices lie in the colored region.

Problem 3.3. Rosencrantz and Guildenstern gamble by repeatedly tossing a biased coin with probability p of heads and probability $q = 1 - p$ of tails, where $0 < p < 1$. Every time the coin comes up heads, Rosencrantz wins a krone from Guildenstern, otherwise Guildenstern wins a krone from Rosencrantz: they play till one of them loses all his money, i.e. 'gets ruined' and so the other 'wins'. Starting with the same sum, say, each with k kroner, they play till one of them is ruined. When is the game expected to be longer: if Rosencrantz wins or Guildenstern?

Problem 3.4 (2022 Bulgarian tournament). A graph G with n vertices is given, and x of its edges are colored red so that each triangle has at most one red edge. The maximum number of vertices in G that induce a bipartite graph equals y . Prove that $n \geq 4\frac{x}{y}$.

Problem 3.5 (Bulgarian autumn competition). A complete bipartite graph $K(A, B)$ is given, such that $|A| = km, |B| = kn; k, m, n \in \mathbb{N}$. (each vertex in A is connected to each vertex in B). The edges of K are colored in k colors. Prove that we can choose a color and a set V of at least $m+n$ vertices such that between any two vertices in V there is a path all edges of which have the chosen color.

Problem 3.6 (IMO Shortlist 2006). Let S be a finite set of points in the plane, no three of which are collinear. For any convex polygon P with vertices from S , let $a(P)$ be the number of vertices of P and $b(P)$ be the number of points of S outside of P . (We consider a line segment to be a convex polygon with two vertices, a point to be a convex polygon with one vertex, and the empty set to be a convex polygon with zero vertices). Prove that for any $x \in (0, 1)$:

$$\sum_P x^{a(P)} (1-x)^{b(P)} = 1$$

where the sum is over all convex polygons P whose vertices are in S .

4 When We Have to Choose

The really interesting use of probability are the problems where you have to define a probability measure yourself. Look at the following beautiful proof by Pach and Toth:

Example 4.1 (Crossing Number Inequality)

Show that any drawing in the plane of a graph with n vertices and $m > 5n$ edges has at least $\frac{4m^3}{243n^2}$ crossings.

Solution. Let a graph G with n vertices and m edges be drawn in the plane with $cr(G)$ crossings. It is a well-known consequence of Euler's formula that any planar graph with v vertices and e edges has $e \leq 3v - 6$. This implies that if a graph has more edges, it must have crossings. Specifically, one can show that for any graph G' , $cr(G') \geq |E(G')| - 3|V(G')|$.

Now, construct a random subgraph G_p of G by selecting each vertex of G independently with probability p . The drawing of G_p is inherited from the drawing of G . Let V_p, E_p be the set of vertices and edges of G_p , and let $X(G_p)$ be the number of crossings in its induced drawing. The expected values are $E[|V_p|] = np$, $E[|E_p|] = mp^2$, and $E[X(G_p)] = cr(G)p^4$. We have the inequality $X(G_p) \geq |E_p| - 3|V_p|$. Taking the expectation of both sides: $E[X(G_p)] \geq E[|E_p| - 3|V_p|]$ $cr(G)p^4 \geq E[|E_p|] - 3E[|V_p|] = mp^2 - 3np$. This

implies $cr(G) \geq \frac{m}{p^2} - \frac{3n}{p^3}$. To maximize the lower bound, we choose p optimally. The right side is maximized for $p = \frac{9n}{2m}$. (The original text had a typo for this value). Let's assume m is large enough that $p \leq 1$. With this choice of p :

$$cr(G) \geq m \left(\frac{2m}{9n} \right)^2 - 3n \left(\frac{2m}{9n} \right)^3 = \frac{4m^3}{81n^2} - \frac{24m^3}{729n^2} = \frac{m^3}{n^2} \left(\frac{36 - 24}{729} \right) = \frac{12m^3}{729n^2} = \frac{4m^3}{243n^2}$$

□

Problem 4.2 (USAMO 2016). A set of 136 points ordered pair are written on a blackboard. For each point (x, y) , at most one of (x, y) and $(-x, -y)$ is written. Alice erases a number of numbers from these pairs(he may remove zero, one or both component of each pair) such that the sum of no two removed numbers is equal to zero. Alice then scores points equal to the number of erased numbers.. What is the maximum score Alice can guarantee?

Problem 4.3 (Sweden 2010). A city has $3n$ citizens. Any two people have at least one common friend. Prove that one can choose n citizen such that any other citizen has at least one friend in this group. can you prove a better bound.

5 Problems

Problem 5.1. Every graph with m edges has an induced bipartite subgraph with at least $m/2$ edges.

Problem 5.2. Let $R(s)$ be the smallest integer such that any 2-coloring of the edges of a complete graph on $R(s)$ vertices contains a monochromatic complete subgraph on s vertices. Prove that $R(s) > 2^{s/2}$.

Problem 5.3. Let $p, q > 0$ be real numbers such that $p + q = 1$. For any positive integers m, n , prove that

$$(1 - p^m)^n + (1 - q^n)^m \geq 1$$

Problem 5.4 (IMO Shortlist 1987). Prove that one can color the set $\{1, 2, \dots, 1987\}$ with four colors such that there is no monochromatic arithmetic progression of length 10.

Problem 5.5 (IMO Shortlist 1989). To each vertex of a regular bipartite graph on $2n$ vertices, a list of more than $\log_2 n$ colors is assigned. Prove that it is possible to choose a color for each vertex from its list such that no two adjacent vertices have the same color.

Problem 5.6 (Russia 2024). In a country with $n > 100$ cities, there are initially no roads. The government announces the cost of building each potential road as a random permutation of the numbers $1, 2, \dots, \frac{n(n-1)}{2}$. The mayor of each city identifies the cheapest road from their city to any other city and builds it. Find the expected number of connected components in the resulting graph.

Problem 5.7. Let Δ be the set of all points (p_1, p_2, p_3) such that p_i are non-negative real numbers and $p_1 + p_2 + p_3 = 1$. Let v_1, v_2, v_3 be three vectors in $\mathbb{R}^3$ such that among them, v_1 has the largest Euclidean norm. If a point (p_1, p_2, p_3) is chosen randomly from Δ with a uniform distribution, prove that for any number c in the interval $(0, \frac{1}{4})$, we have

$$\mathbb{P}(\|p_1 v_1 + p_2 v_2 + p_3 v_3\| \leq c \|v_1\|) \leq 4c.$$

Problem 5.8. Let $A = A_1 \times A_2 \times \dots \times A_n$, where the A_i are finite sets. A "brick" is a subset of A of the form $B_1 \times \dots \times B_n$ where each B_i is a proper subset of A_i . Prove that A cannot be covered by fewer than 2^n bricks.

Problem 5.9. If $\{(A_i, B_i) : 1 \leq i \leq m\}$ is a cross-intersecting family (of pairs of finite sets) then

$$\sum_{i=1}^m \left(\frac{|A_i| + |B_i|}{|A_i|} \right)^{-1} \leq 1, \quad (36)$$